

Exploring and Analyzing University Student Performance

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STAT 419: Applied Multivariate Statistics

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Executive Summary

The purpose of this study was to explore how student habits relate to academic outcomes. Specifically, we examined whether gender and academic major jointly influence students' final GPA and class attendance, and we investigated which behavioral variables explain the variation in student performance. Finally, we explored whether student habits could distinguish groups of students by gender, age, or major.

The analysis used the University Student Performance & Habits Dataset from Kaggle, which includes data from +5k students. Each observation represents one student and includes demographic variables (gender, age, and major), academic outcomes (final GPA and attendance percentage), and habit measures (study hours, sleep hours, and weekly social hours).

Several multivariate statistical techniques were used to analyze the data. The results of a Two-Way MANOVA showed a significant interaction between gender and major on the joint outcomes of final GPA and attendance ($p = 0.041$). This means that the relationship between gender and academic performance differs depending on a student's major, but the individual effects of gender and major were not statistically significant. Exploratory analysis using PCA revealed that class attendance explained the majority of variation in student habits, about 90% to be precise. This indicated that attendance dominated in explaining differences in student performance patterns. Finally, discriminant analysis showed no statistically significant variables that clearly separate students by gender, age, or major. This suggests that behaviors such as study time, sleep, and social activity are relatively similar across these demographic groups.

Overall, the results indicate that attendance is the strongest factor associated with differences in student performance patterns, while demographic characteristics alone do not strongly distinguish student behavior and final GPA.

Main Report of University Student Performance

Introduction & Data Description

Throughout the quarter in this course, STAT 419: Applied Multivariate Statistics, we have learned many methods and tests that aim to understand multivariate data and analyze patterns within the data. This project aims to apply the multivariate statistical methods learned in the course. We will use methods such as Two-Way MANOVA, Discriminant Analysis, and Principal Component Analysis in order to ask the research questions we have gathered in the following data.

The reason we chose the dataset, University Student Performance & Habits Dataset, taken from [Kaggle](#) (Robiul Hasan Jisan), is to analyze habits that we might relate to as college students. The dataset contains information about student demographics, study habits, attendance, and their GPA's as the variable of interest. Each row is one student. There are 10 columns in total, and the data has already been cleaned for analysis, with no missing or inconsistent values. The direct source of the data has been removed for privacy reasons, but 5000 individual observations were taken from students across their respective universities. We wanted to analyze these habits to see their combined effects on final GPA grades that we, college students, focus on during our academic careers.

Here is a brief description of the data's variables and what they represent:

- Gender: Gender of the student (Male/Female), Categorical
- Age: Age of the student (18–24 years), Integer
- Major: Field of study (Computer Science, Engineering), Categorical
- Attendance_Pct: Percentage of classes attended (0–100%), Float

- Study_Hours_Per_Day: Average hours spent studying per day, Float
- Previous_CGPA: GPA from the previous semester (0.0–4.0 scale), Float
- Sleep_Hours: Average hours of sleep per night, Float
- Social_Hours_Week: Average hours spent socializing/partying per week, Integer
- Final_CGPA: Target Variable: final cumulative GPA (0.0–4.0 scale)

The following research questions will guide our analysis and tests.

- Are there gender and major effects, as well as an interaction effect, jointly on both students' final GPA and attendance percentage?
- Which variables (attendance, study hour, sleep hour, social hour) explain the total variation in the data?
- Which variables (attendance, study hour, sleep hour, social hour) best distinguish the groups

All analyses will be done in R, and some functions used will be accessed from the R Misc. Code sections of the STAT 419 Modules tab in Canvas.

Research Question #1: Are there gender and major effects, as well as an interaction effect, jointly on both students' final GPA and attendance percentage?

For our first research question, we want to test whether gender, major, and their interaction influence the combination of students' final GPA and attendance percentage simultaneously. In order to do that, we decided to use a Two-Way MANOVA test since we are testing two categorical variables' effect on multiple response variables. This test is better than running multiple ANOVA tests because we also want to test the interaction effects of gender and major on the response variables, which can help build a better explanation of the variation within

the data. We also benefit from using a Two-Way MANOVA test, as opposed to multiple separate ANOVA tests, because we can keep the Type 1 error from being inflated. The following test will be a Roy's Largest Root Test with the following parameters and hypotheses:

Factor A (Gender):

$$H_0: \alpha_i = \mathbf{0} \text{ for } i = 1, 2$$

$$H_A: \alpha_i \neq \mathbf{0} \text{ for some } i$$

$$i \in \{1 = \text{Male}, 2 = \text{Female}\}$$

Factor B (Major):

$$H_0: \beta_j = \mathbf{0} \text{ for } j = 1, 2, 3, 4, 5, 6$$

$$H_A: \beta_j \neq \mathbf{0} \text{ for some } j$$

$$j \in \{1 = \text{Engineering}, 2 = \text{Business}, 3 = \text{Mathematics}, 4 = \text{Economics}, 5 = \text{Economics}, 6 = \text{Computer Science}\}$$

Interaction A x B (Gender x Major):

$$H_0: \gamma_{ij} = \mathbf{0} \text{ for all } i \text{ and } j$$

$$H_A: \gamma_{ij} \neq \mathbf{0} \text{ for some } i \text{ and } j$$

Where:

α_i = the effect of the i-th level of Gender jointly on Final_CGPA and

Attendance_Pct

β_j = the effect of the j-th level of Major jointly on Final_CGPA and

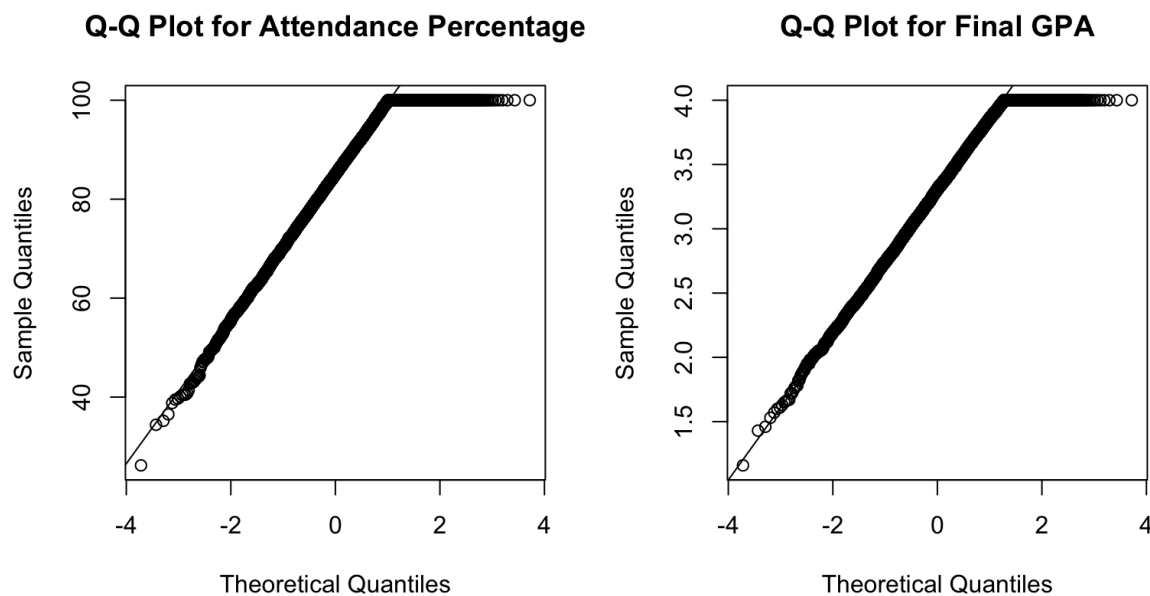
Attendance_Pct

γ_{ij} = the interaction effect of the ij-th combination of Gender and Major jointly on

Final_CGPA and Attendance_Pct

For a MANOVA test to be conducted, we need to validate some assumptions before we test. The first assumption is independence of observations. We can conclude that this assumption was not violated because the data description explained that the data gathered was independent. The second assumption is multivariate normality within the data. To test that, used Q-Q plots for plotting the data and analyzing its normality.

Figure 1: Q-Q Plots



Attendance_Pct: The center of the distribution follows the line well. The right tail flattens near 100% attendance, because the variable is bounded at 100.

Final_CGPA: Most of the points lie very close to the straight line. Only the extreme right tail flattens slightly near 4.0, which happens because GPA is bounded above at 4.0.

This type of deviation in the right tail is expected with bounded variables and does not indicate a serious violation. The plots show a strong linear pattern in the middle and mild tail compression due to natural variable limits, visually indicating that our data satisfies the

approximate normality assumption. Given the large sample size and balanced group sizes, these minor departures from normality are unlikely to substantially affect the MANOVA results.

The third and final assumption is the equal variance assumption between the covariance matrices in the data. We will test that using a Box's M-Test with the following parameters and Hypothesis:

$$H_0: \Sigma_1 = \Sigma_2 = \dots = \Sigma_{12}$$

H_A : At least one covariance matrix differs

where the covariance matrices correspond to the Gender x Major groups for the response variables Final_CGPA and Attendance_Pct.

With test statistics $u = 27.24$ and $p = 0.7489$, we fail to reject H_0 since our p-value, 0.7489, is lower than our confidence level of 0.05. At the $\alpha=0.05$, there is no evidence that the covariance matrices differ across the Gender $\times$ Major groups. Therefore, the MANOVA assumption of equality of covariance matrices for Final_CGPA and Attendance_Pct is satisfied, and the MANOVA results can be considered valid.

For the MANOVA test, we will use Roy's Test. Roy's test examines whether the largest eigenvalue of the hypothesis matrix relative to the error matrix shows a significant difference between group mean vectors of Final GPA and Attendance Percentage. After conducting this test in R we get the following result.

```

              Df      Roy approx F num Df den Df  Pr(>F)
as.factor(Gender)      1 0.00040256   1.0038     2   4987 0.36657
as.factor(Major)       5 0.00148427   1.4807     5   4988 0.19253
as.factor(Gender):as.factor(Major)  5 0.00232643   2.3209     5   4988 0.04079 *
Residuals              4988
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Looking at the eigenvalues in the output, Roy's largest root ($\lambda_1=0.00232643$) was much larger than the remaining eigenvalues, indicating that the group means differences occur

primarily along one dimension. This suggests that the mean vectors are approximately collinear, making Roy's Largest Root the most appropriate MANOVA test statistic.

Since the p-value for the interaction effect is $0.041 < 0.05$, and its F statistic is $F(5,4988) = 2.32$, we reject the null hypothesis for the interaction effect and do not proceed to test the main effects. At the 0.05 level, we can conclude that there is a significant interaction effect of Gender and Major jointly on Final CGPA and Attendance Pct.

MANOVA assumes multivariate normality of the response variables within groups, independence of observations, and homogeneity of covariance matrices across groups. Because the dataset contains a large sample size ($n \approx 5000$), and the Q-Q plots did not signal any strong non-linearity patterns, the multivariate normality assumption is not violated. Independence is assumed since each observation corresponds to a different student in the sample. Each student, according to the data description, was sampled independently. Homogeneity of covariance matrices was tested using the Box's M test and found not to violate the assumption.

To answer the research question posed in the beginning, we can say that there does exist an interaction effect of gender and major on attendance percentage and final GPA scores jointly. That is, the effect of gender or major on attendance and final GPA depends on the other variable as well.

Research Question #2: Which variables (attendance, study hour, sleep hour, social hour) explain the total variation in the data?

While we have a statistically significant interaction effect between gender and major, we also want to analyze which variable accounts for the majority of the variance in the data. A good way to carry this out is using PCA (Principal Component Analysis). We decided to use PCA to

see which of our habit variables accounts for the most variation in the data. For PCA, we will use the 4 quantitative variables: attendance, study hours, sleep, and social time as explanatory variables for the final GPA.

First, we will inspect the coefficients of the principal components (loadings) to get an idea of how the distribution of GPA is formed. Then we will look into four different ways to figure out how many principal components to keep.

Figure 2

Loadings:				
	Comp.1	Comp.2	Comp.3	Comp.4
Attendance	1.000	0.003	0.005	0.002
Study	0.005	-0.012	-1.000	0.017
Sleep	-0.002	-0.005	0.017	1.000
Social	0.003	-1.000	0.012	-0.005

	Comp.1	Comp.2	Comp.3	Comp.4
SS loadings	1.00	1.00	1.00	1.00
Proportion Var	0.25	0.25	0.25	0.25
Cumulative Var	0.25	0.50	0.75	1.00

The first PC and second PC can be interpreted using their loadings. Looking at Figure 2 above, attendance dominates the first PC, so it is weighted the most and creates the size of the GPA distribution. Attendance and social contrasts the most in the second PC, suggesting that attendance and social causes the most change in the GPA distribution.

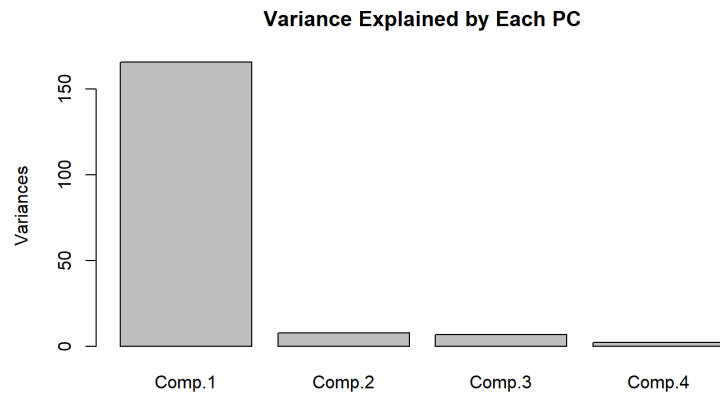
We will analyze the importance of each of the components and their proportion in the variance to see which of the components explains the variance the most.

Figure 3

Importance of components:				
	Comp.1	Comp.2	Comp.3	Comp.4
Standard deviation	12.8668197	2.79735540	2.56739043	1.4234921
Proportion of Variance	0.9096528	0.04299605	0.03621738	0.0111338
Cumulative Proportion	0.9096528	0.95264882	0.98886620	1.0000000

From an initial inspection of the variables and their contributions to the total variation, it looks like the first principal component explains 90% of the total variation. The standard deviation of the first principal component is also much higher than the rest. To fully understand the weight of each PC, we used a Scree Plot to plot the variance for each of the PC's.

Figure 2



Our scree plot supports our findings from the statistics above. One PC indeed is dominating. We can also choose the corresponding eigenvalues that are greater than the average of the eigenvalues. First, the first eigenvalue 12.9^2 is greater than 45.5, which is the average value of the eigenvalues.

Last, we will perform the Principal Components Significance Test. We will add 1 additional principal component and see if its eigenvalues are equal, and stop until one is significantly different from the rest. This test typically yields conservative results, meaning that it tends to suggest keeping more principal components than needed.

	Eigenvalue	k	u	df	p-value
Comp.1	165.555049	4	27543.285	9	0
Comp.2	7.825197	3	2271.118	5	0
Comp.3	6.591494	2	1645.780	2	0

Because our p-values of 0 are less than .05 for all the principal component comparisons, we reject H_0 , and we have convincing evidence that all four principal components are significant and should be retained to describe the variation in attendance rate, hours of sleep, hours of study, and hours of socializing.

From the scree plot and standard deviations of the eigenvalues, we should keep one PC. But we should confirm our guess that there is one dominating variable. It could be the case that the variables are correlated, or one variable has much higher variation than the rest. This will cause one PC to dominate the rest. Let's take a look at the relationship between the variables.

	Attendance	Study	Sleep	Social
Attendance	165.583	0.750	-0.316	0.411
Study	0.750	6.595	-0.080	0.017
Sleep	-0.316	-0.080	2.029	0.027
Social	0.411	0.017	0.027	7.828
	Attendance	Study	Sleep	Social
Attendance	1.000	0.023	-0.017	0.011
Study	0.023	1.000	-0.022	0.002
Sleep	-0.017	-0.022	1.000	0.007
Social	0.011	0.002	0.007	1.000

The covariance matrix shows us the variance and covariance. The diagonal values show us each variable's variance, and the off-diagonal values are the covariance of the two variables. Attendance varies a lot more than the other variables. The off-diagonal values are all small, so this suggests that the variables don't affect each other much.

The correlation matrix shows us how much each variable relates to the others. The diagonal values show us the relationship between a variable and itself, so it is always 1. The

off-diagonal values are the part we are interested in. From the matrix, it doesn't look like there are any significant relationships between any of the variables.

In summary, we have one variable that has much higher variance than the rest; therefore, one principal component will dominate the rest. We will move on with our analysis to figure out how many variables we need.

After analyzing the different components of PCA, we think it is most appropriate to keep 1 variable for inference. We should also note that attendance is creating most of the variation in GPA. While this is true, we can also use the results from the Two-Way MANOVA that indicate an interaction effect to truly understand the patterns in the data.

Research Question #3: Which variables (attendance, study hour, sleep hour, social hour) best distinguish the groups?

While PCA showed us that attendance was a big explainer in the variation across the data, we also want to see how we can group certain students to predict what their outcome will be. We will test whether the group of variables within the data are significant in separating Age, Gender and Major groups within these college students.

Moreover, with only two groups within the Gender group (Male and Female) there isn't possibly a way to plot the discriminant functions since discriminant functions are made with the number of groups - 1.

Gender

Using R, we tested the discriminant tests of gender which yielded a test statistic of 0.2296. At the .05 significance level, we do not have sufficient evidence that Sleep Hours, Previous GPA, Social Hours per Week, Final CGPA, Attendance Percentage, and Study Hours

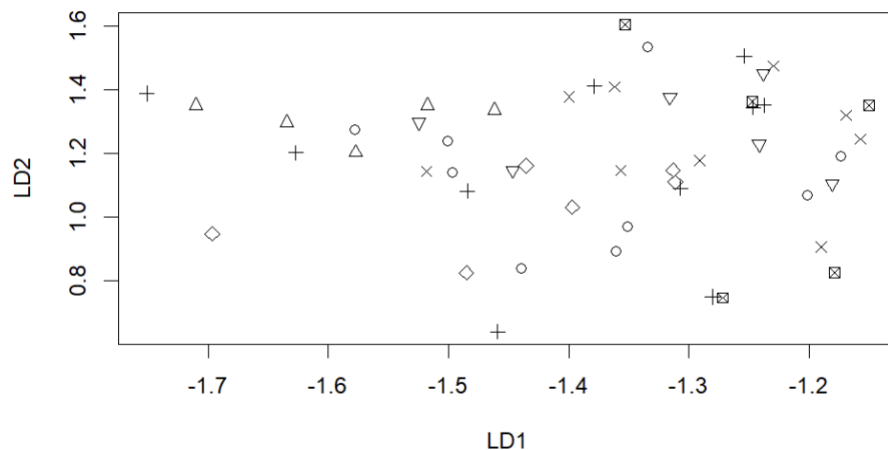
per Day collectively help distinguish between Male and Female students, after adjusting for the other variables in the model. We cannot, however, conclude that these variables have no relationship with gender grouping, nor can we draw broader conclusions about intelligence from this result.

Age

Moving on to Age, we did the same tests, which also yielded p-values of above 0.05. At the .05 level, we cannot conclude that Sleep_Hours, Previous_GPA, Social_Hours_week, Final_CGPA, Attendance_Pct, and Study_Hours_Per_Day do not contribute and are not useful for separating the groups of ages (18 - 24), while adjusting for the other variables in the model. Important to note from this test is that this shows that age also does not discriminately show that there is a separation of intelligence based on the person's age. Without any significant variable that separates the age groups, we can conclude from the results of this test that there are no specific areas that make one group separate from another depending on their age.

For this variable, we can actually plot the different groups, which helps us identify the lack of possible groups that could be made. In the following plot, we can see that there is no distinguishable group between the age groups, which was previously confirmed by the significance tests

Figure 3: Age Discriminant Plot



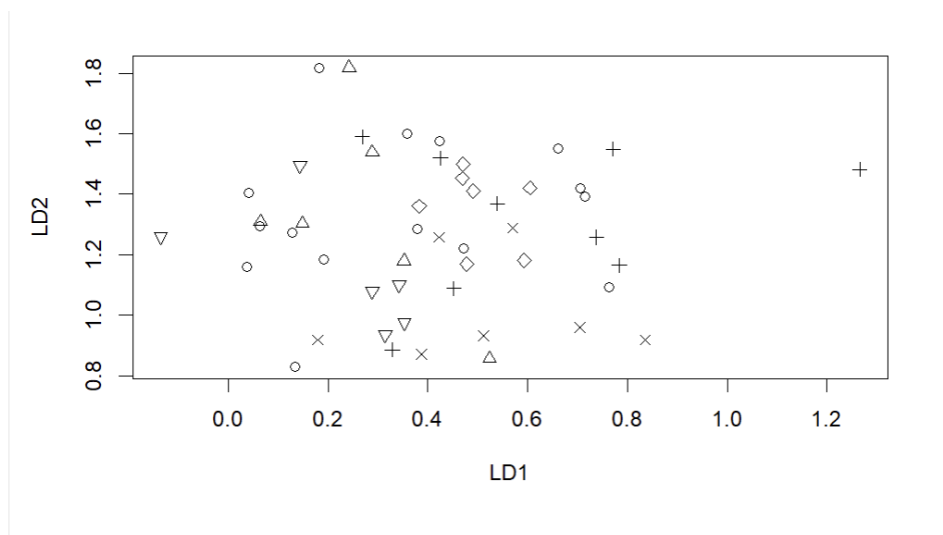
Major

Finally, we will test whether majors are significant enough to linearly separate the data from the given dataset. After another test, we also received p-values much over .50. At .05 level, we cannot conclude that Sleep_Hours, Previous_GPA, Social_Hours_week, Final_CGPA, Attendance_Pct, and Study_Hours_Per_Day do not contribute and are not useful for separating the groups of Engineering, Business, Mathematics, Economics, Psychology, Computer Science, while adjusting for the other variables in the model.

It is important to note what our data reveals. Because there isn't a significant discriminant function or significant variable that separates our groups from one another, we can claim that despite a person's major there isn't any reason a person with a Business degree should be performing better than someone with a Computer Science major. Therefore, their major does not predict their performance (Sleep_Hours, Previous_GPA, Social_Hours_week, Final_CGPA, Attendance_Pct, and Study_Hours_Per_Day).

We can also see this pattern of non-linearly separable data in the following plot.

Figure 4: Major Discriminant Plot



Appendix

Appendix A: R Code for Statistical Analyses

A1. Data Preparation

```
library(dplyr)

Student_data <- read.csv("Student_data.csv")

df_Performance <- Student_data |>
  select(Gender, Major, Attendance_Pct, Final_CGPA) |>
  mutate(
    Gender = recode(Gender,
      "Male" = 1,
      "Female" = 2),
    Major = recode(Major,
      "Engineering" = 1,
      "Business" = 2,
      "Mathematics" = 3,
      "Economics" = 4,
      "Psychology" = 5,
      "Computer Science" = 6)
  )
```

A2. Two-Way MANOVA

```
performance.manova <- manova(
  cbind(Final_CGPA, Attendance_Pct) ~
  as.factor(Gender) * as.factor(Major),
  data = df_Performance
)

summary(performance.manova, test = "Roy")
```

A3. Normality Diagnostics

```
par(mfrow = c(1,2))

qqnorm(df_Performance$Attendance_Pct,
  main = "Q-Q Plot for Attendance Percentage")
qqline(df_Performance$Attendance_Pct)
```



```
qqnorm(df_Performance$Final_CGPA,
       main = "Q-Q Plot for Final GPA")
qqline(df_Performance$Final_CGPA)
```

```
par(mfrow = c(1,1))
```

A4. Principal Components Analysis

```
Student_matrix <- as.matrix(Student_data)
```

```
Student_matrix <- Student_matrix[, -c(1:4,7)]
```

```
Student <- apply(Student_matrix, 2, as.numeric)
```

```
colnames(Student) <- c(
  "Attendance",
  "Study",
  "Sleep",
  "Social",
  "GPA"
)
```

```
pc_Student <- princomp(Student[1:5000, 1:4])
summary(pc_Student)
```

A5. Scree Plot

```
screeplot(pc_Student,
          main = "Variance Explained by Each PC")
```

A6. PCA Significance Test

```
princomp.test <- function(pr.out){
```

```
  lambda <- pr.out$sdev^2
  n <- pr.out$n.obs
  p <- length(lambda)
```

```
  mult <- n - ((2*p + 11)/6)
```

```
  u <- matrix(0,p-1)
  df <- matrix(0,p-1)
```

```

p.val <- matrix(0,p-1)

for(k in 2:p){

  l.bar <- mean(lambda[(p-k+1):p])

  u[k-1] <- mult * (k*log(l.bar) -
    sum(log(lambda[(p-k+1):p])))

  df[k-1] <- .5*(k-1)*(k+2)

  p.val[k-1] <- 1 - pchisq(u[k-1], df[k-1])
}

out <- cbind(lambda[1:(p-1)],
  p:2,
  rev(u),
  rev(df),
  rev(p.val))

dimnames(out)[[2]] <- c(
  "Eigenvalue", "k", "u", "df", "p-value"
)

return(out)

}

princomp.test(pc_Student)

```

A7. Box's M Test for Equality of Covariance

```

Box.M <- function(samp.covs, n){

  k <- length(n)
  pp <- dim(samp.covs[[1]])

  df <- n - 1

  S.dets <- sapply(samp.covs, det)

  Sp <- matrix(0, nrow = pp[1], ncol = pp[1])

  for (i in 1:k){

```

```

    Sp <- Sp + df[i] * samp.covs[[i]]
  }

  Sp <- Sp / sum(df)

  det.Sp <- det(Sp)

  logM <- sum(df/2 * log(S.dets)) -
    (sum(df)/2) * log(det.Sp)

  c <- (sum(1/df) - 1/sum(df)) *
    ((2*pp[1]^2 + 3*pp[1] - 1) /
    (6*(pp[1] + 1)*(k - 1)))

  u <- -2*(1-c)*logM

  u.df <- .5*(k-1)*pp[1]*(pp[1]+1)

  p.value <- 1 - pchisq(u, u.df)

  list("u" = u,
       "p.value.u" = p.value,
       "chi.df" = u.df)
}

```

Appendix B: Selected Statistical Output

Two-Way MANOVA (Roy's Largest Root)

```

              Df      Roy approx F num Df den Df  Pr(>F)
as.factor(Gender)      1 0.00040256   1.0038     2   4987 0.36657
as.factor(Major)       5 0.00148427   1.4807     5   4988 0.19253
as.factor(Gender):as.factor(Major)  5 0.00232643   2.3209     5   4988 0.04079 *
Residuals              4988
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

PCA Loadings

Loadings:

	Comp.1	Comp.2	Comp.3	Comp.4
Attendance	1.000	0.003	0.005	0.002
Study	0.005	-0.012	-1.000	0.017
Sleep	-0.002	-0.005	0.017	1.000
Social	0.003	-1.000	0.012	-0.005

	Comp.1	Comp.2	Comp.3	Comp.4
SS loadings	1.00	1.00	1.00	1.00
Proportion Var	0.25	0.25	0.25	0.25
Cumulative Var	0.25	0.50	0.75	1.00

PCA Components

	Eigenvalue	k	u	df	p-value
Comp.1	165.555049	4	27543.285	9	0
Comp.2	7.825197	3	2271.118	5	0
Comp.3	6.591494	2	1645.780	2	0

Covariance Matrix

	Attendance	Study	Sleep	Social
Attendance	165.583	0.750	-0.316	0.411
Study	0.750	6.595	-0.080	0.017
Sleep	-0.316	-0.080	2.029	0.027
Social	0.411	0.017	0.027	7.828
	Attendance	Study	Sleep	Social
Attendance	1.000	0.023	-0.017	0.011
Study	0.023	1.000	-0.022	0.002
Sleep	-0.017	-0.022	1.000	0.007
Social	0.011	0.002	0.007	1.000

Appendix C: Figures

Figure 1: Q-Q Plots

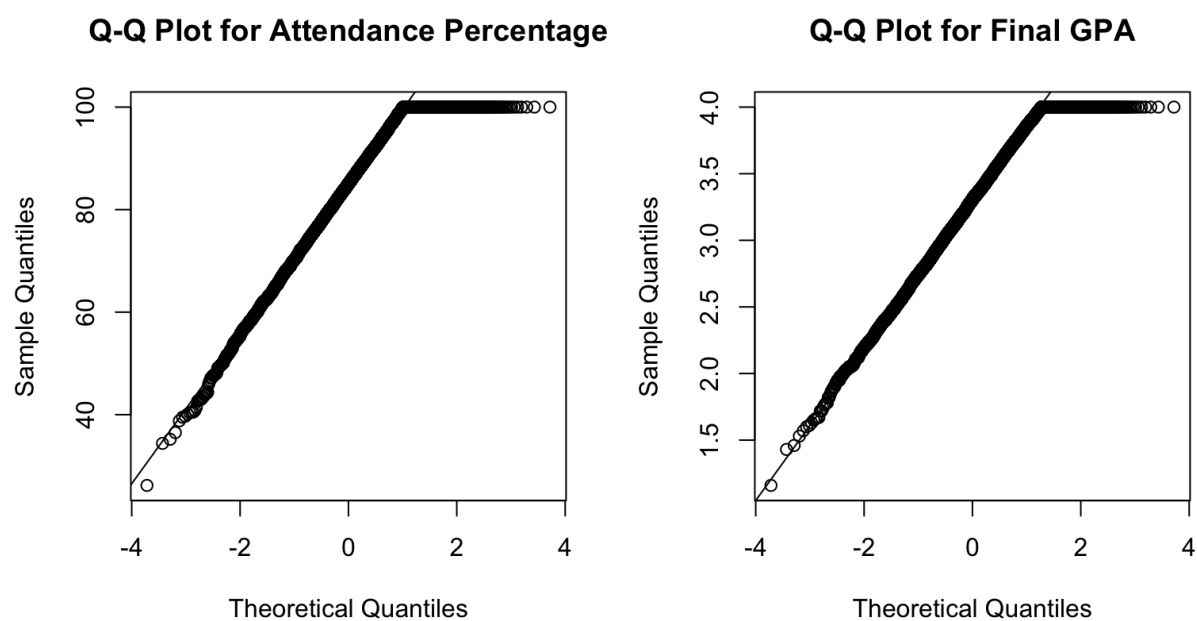
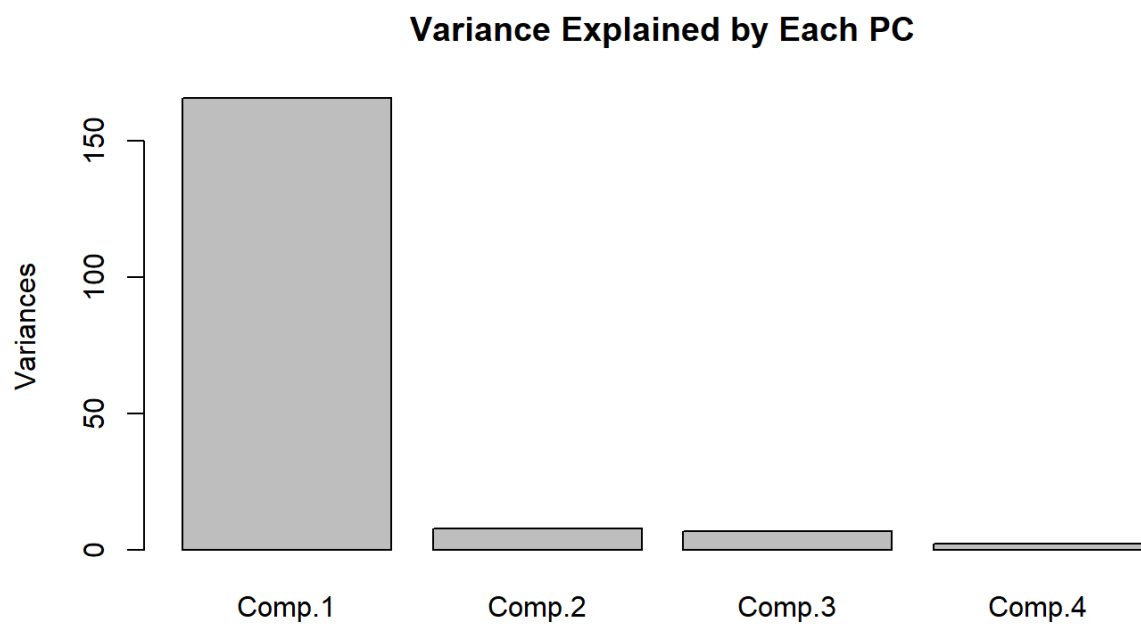
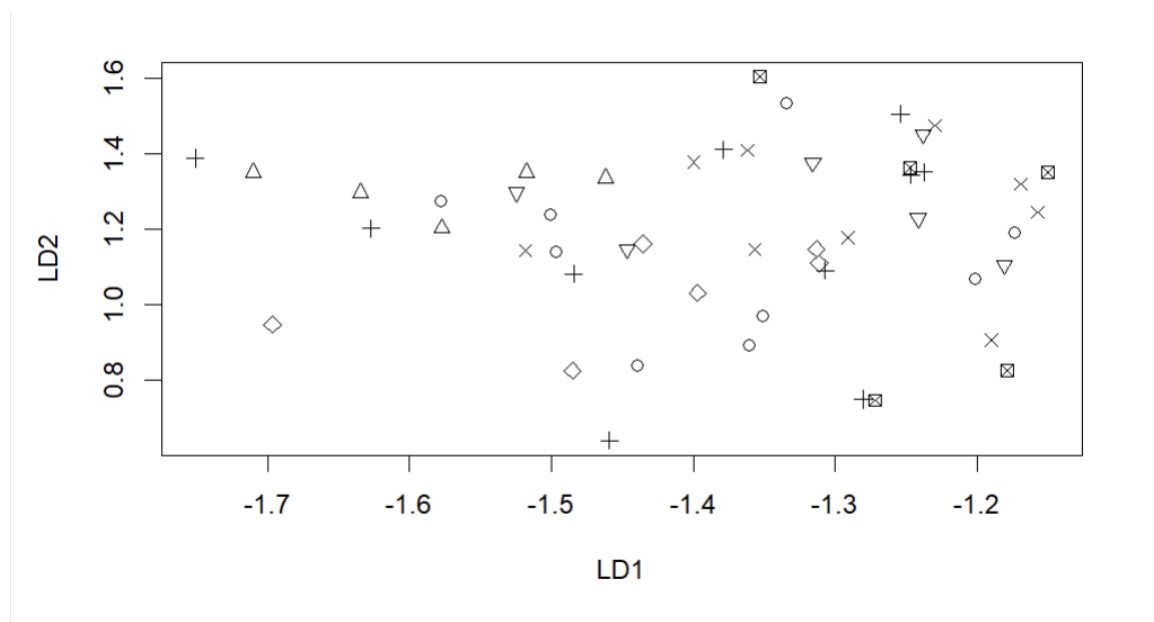


Figure 2: Principal Component Variance Plot**Figure 3: Age Discriminant Plot****Figure 4: Major Discriminant Plot**

